Step 4 : **Test statistic**

The test statistic under H_0 is

$$Z = \frac{\overline{X} - \overline{Y}}{\sqrt{\frac{S_X^2}{m} + \frac{S_Y^2}{n}}}.$$

The sampling distribution of Z under H_0 is the $N(0,1)$ distribution.

Step 5 : **Calculation of the Test Statistic**

The value of Z is calculated for the given sample informations from

$$z_0 = \frac{\overline{x} - \overline{y}}{\sqrt{\frac{s_X^2}{m} + \frac{s_Y^2}{n}}} \quad \text{as}$$

$$z_0 = \frac{50 - 51}{\sqrt{\frac{4^2}{100} + \frac{5^2}{150}}}$$

Thus, $z_0 = -1.75$

Step 6 : **Critical value**

Since H_1 is a two-sided alternative, the critical value at 5% level of significance is $z_e = z_{0.025} = 1.96$.

Step 7 : **Decision**

Since H_1 is a two-sided alternative, elements of the critical region are determined by the rejection rule $|z_0| \geq |z_0|$. Thus it is a two-tailed test. But, $|z_0| = 1.75$ is less than the critical value $z_e = 1.96$. Hence, it may inferred as the given sample information does not provide sufficient evidence to reject H_0. Therefore, it may be decided that there is no sufficient evidence in the given sample to conclude that performance of boys and girls in the Model Examination conducted in the subject of Statistics differ significantly.

1.13 TEST OF HYPOTHESES FOR POPULATION PROPORTION

Procedure:

Step 1 : Let P denote the proportion of the population possessing the qualitative characteristic (attribute) under study. If p_0 is an admissible value of P, then frame the null hypothesis as $H_0: P = p_0$ and choose the suitable alternative hypothesis from

(i) $H_1: P \neq p_0$ (ii) $H_1: P > p_0$ (iii) $H_1: P < p_0$

Step 2 : Let p be proportion of the sample observations possessing the attribute, where n is large, $np > 5$ and $n(1 - p) > 5$.

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